

High-Frequency MPCs and the Transmission of Policy*

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Abstract

Marginal propensities to consume (MPCs) are central calibration targets in heterogeneous-agent New Keynesian (HANK) models. Using monthly records from an Indian household panel, we examine how measurement frequency affects MPC estimates. Pooled transitory income pass-through rises from 0.028 monthly to 0.088 at four months and 0.248 biennially. This gradient persists across consumption definitions, weighting, trimming, and pandemic exclusions. In India-calibrated HANK models, alternative MPC targets produce large differences in fiscal transmission, including an eightfold difference in cumulative consumption responses in the aggregate-liquidity design. Monetary responses vary less. Measurement frequency therefore matters for both MPC estimation and policy calibration.

Keywords: Marginal propensity to consume; observation frequency; income dynamics; measurement error; heterogeneous-agent New Keynesian models.

JEL classification: E21, D12, C81, E52, E62.

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1 Introduction

Heterogeneous-agent New Keynesian (HANK) models have become an important framework for studying monetary and fiscal policy (Kaplan et al., 2018; Auclert et al., 2021). Their quantitative predictions depend on how households respond to changes in disposable income, making the marginal propensity to consume (MPC) a central calibration target. An influential source of these targets is the Panel Study of Income Dynamics (PSID), whose biennial observations have been used to estimate consumption responses to income shocks (Kaplan et al., 2014). These estimates often discipline models operating at a quarterly frequency. This creates a measurement question with direct consequences for policy analysis: does observing households only once every two years change the estimated MPC, and hence the consumption response that the model is calibrated to reproduce?

We study this question using the Consumer Pyramids Household Survey (CPHS), an Indian panel that records income and consumption month by month through interviews conducted every four months. The monthly records allow us to construct panels at monthly, quarterly, wave, annual, and biennial frequencies and apply a common estimation framework to each. We can therefore examine how MPC estimates change with the measurement frequency within one data source, rather than relying on comparisons across surveys or countries. The differences are large: in our pooled monthly, wave, and biennial comparisons, the estimated transitory income pass-through rises from 0.028 to 0.088 to 0.248, respectively. We then use these estimates to calibrate HANK models of India and quantify how the choice of measurement frequency changes their predictions for monetary and fiscal transmission.

The paper builds on two approaches to measuring consumption responses. Studies of observed income changes, including lottery prizes, estimate how households spend particular windfalls (Fagereng et al., 2021). The approach developed by Blundell et al. (2008), hereafter BPP, instead uses the joint variation in income and consumption to estimate the transmission of permanent and transitory income shocks. Estimates from these approaches need not describe the same response: they differ in the income variation used for identification, the measurement horizon, and the assumptions connecting observed data to household behavior. Crawley (2020) shows that accounting for time aggregation can substantially change covariance-based estimates, while Commault (2022) shows that relaxing assumptions about consumption dynamics can reconcile part of the difference between semistructural and natural-experiment estimates. We complement this work by examining estimates across several frequencies within one source of household data.

Our empirical analysis uses observations from 2014–2025. We first remove variation in log income and log consumption associated with demographic characteristics and year–month effects. Our baseline aggregation method multiplies the mean monthly residual within a period by the number of months in the target window. We examine alternative constructions in the robustness analysis. At each frequency, households must have valid observations in at least three consecutive periods. The estimation samples can therefore differ across frequencies, even though they originate in the same monthly panel.

We begin by estimating consumption responses to permanent and transitory income shocks at all five frequencies. The permanent-income point estimates increase more strongly with the measurement window than the transitory-income estimates. Precision also differs across frequencies: the monthly, quarterly, and wave estimates in this comparison are imprecise, while the annual and biennial estimates are more precisely estimated. These results motivate a closer examination of the monthly, wave, and biennial comparisons, including how their patterns differ across household groups.

In the household-group analysis, differences in point estimates are small at the monthly frequency and become more pronounced at the wave frequency. The wave MPC is 0.160 for poor hand-to-mouth (HtM) households, 0.100 for wealthy HtM households, and 0.073 for non-HtM households. The baseline biennial estimates retain this ordering, although the poor HtM estimate is substantially less precise. We assess differences across frequencies using 500 bootstrap replications with common household draws, accounting for dependence between the estimates. Eleven of the twelve group-specific and pooled comparisons reject equality at the 5 percent level. The exception is the wave–biennial comparison for poor HtM households.

The pooled frequency gradient persists across the reported consumption definitions, weighting choices, outlier rules, and state-by-period demeaning of growth rates. These choices affect the levels of the estimates, sometimes substantially, but preserve their monthly–wave–biennial ordering. The pattern also survives removing calendar years 2020 and 2021 entirely, including interview waves and biennial growth intervals that overlap those years. The resulting pooled estimates are 0.019 monthly, 0.080 at the wave frequency, and 0.242 biennially. The monthly–biennial gap is then 0.223, compared with 0.220 in the baseline. Thus, the pooled pattern does not depend on including the pandemic years. Household-group rankings are less stable across specifications, and the demanding exclusions reduce precision in some subgroup comparisons.

We next examine the policy consequences of using these estimates as calibration targets. We solve quarterly one-asset HANK models using the sequence-space method of [Auclert et al. \(2021\)](#). One calibration design adjusts aggregate liquid asset supply to match the pooled MPC. The other calibrates type-specific discount factors to match the group-level MPCs, with household-type shares taken from Gupta, Singh, and Pizzolon. These externally calibrated shares remain constant across the frequency comparisons. Both designs use a common baseline income process, allowing us to isolate the consequences of changing the MPC targets.

The predicted consumption response to a deficit-financed transfer is highly sensitive to this choice. When empirical estimates are assigned directly to quarterly MPC targets, the aggregate-liquidity design produces an approximately eightfold difference in the cumulative twelve-quarter consumption response between the monthly-target and biennial-target calibrations. The monthly target lies below the minimum MPC attained within the calibration search, so its reported policy response comes from a boundary economy rather than an exact target match. The three-type design produces the same qualitative fiscal sensitivity using

the group estimates directly. Monetary responses vary less across calibrations. This relative stability is conditional on the model’s asset structure, income process, and policy rules, rather than a general result about monetary transmission.

Mapping an empirical estimate into the model’s quarterly time step introduces a further choice. Alongside the direct mapping, we consider a conversion that interprets an estimate over a longer window as a cumulative response generated by a constant monthly spending hazard. The two mappings can produce different rankings of fiscal responses. The conversion is an additional economic assumption, not an identity for window-averaged data. The policy exercise therefore depends both on the empirical estimate selected and on what that estimate is assumed to measure.

The remainder of the paper proceeds as follows. We first describe the data and sample construction, then present the estimation framework and temporal aggregation methods. We report the full-sample and household-group estimates, followed by the cross-frequency tests and robustness exercises. We then examine policy transmission in the India-calibrated HANK models and conclude.

2 Data: The Consumer Pyramids Household Survey

CPHS, run by the Centre for Monitoring Indian Economy, is a nationally representative rotating panel that has interviewed roughly 170,000 households three times a year since 2014 ([Centre for Monitoring Indian Economy, 2025](#)). Each interview (a “wave”) collects the household’s total income and roughly 150 expenditure items *for each of the four preceding months*, so a fully observed household contributes twelve household-month observations per year.

The monthly recall design is both the survey’s great advantage and its central hazard. It is what allows us to estimate the same moment at three frequencies; it is also what makes income measured within one interview sitting mechanically different from income measured across sittings.

2.1 Cleaning protocol and sample

We keep household-months with an accepted consumption interview, a positive revised country-level survey weight, positive income and consumption, and at least 24 usable months per household.

Consumption is the sum of eleven nondurable and service expenditure heads chosen to match the definition in [Kaplan et al. \(2014\)](#): food, intoxicants, restaurants, toiletries, power and fuel, transport, communication, recreation, clothing and footwear, health, and education. Robustness variants drop health and education (the BPP nine-head basket), use CMIE’s

total expenditure, or restrict to food. Income is total household income from all sources. Both are deflated by the all-India CPI (2012 base) and divided by household size, held fixed at its within-household mean to prevent spurious re-ranking. Growth rates are trimmed at the survey-weighted 1st and 99th percentiles within each frequency; Section 4.2 shows fixed KVV bounds and no trimming as alternatives. Survey weights are the revised country-level household weights: the month-specific weight for the monthly and biennial panels, and the interview-level weight (the maximum reported value within the household-wave, of which there is one positive value per wave) for the wave panel; weights are normalized to mean one within each panel.

Household Income: We construct monthly household income using the income module of the Consumer Pyramids Household Survey (CPHS). Unlike a conventional questionnaire in which respondents provide answers to a fixed sequence of narrowly defined questions, the CPHS interview is conducted in a conversational format, with interviewers using probing and recall prompts and, where necessary, consulting other household members to elicit information on different sources of income. The income module records income for the month preceding the interview and the three preceding months (Month L, Month L-1, Month L-2, and Month L-3), covering sources such as rent, business profits, private and government cash transfers, in-kind transfers, imputed income, proceeds from asset sales, and lottery or gambling income. We aggregate the reported amounts across all income sources and household members to construct total monthly household income, which represents the total income accruing to the household during the reference month.

Household consumption; We construct monthly household consumption using the consumption module of the Consumer Pyramids Household Survey (CPHS). As with the income module, the CPHS does not rely on a single, narrowly defined consumption question; rather, expenditure information is elicited through a conversational interview, with interviewers using probing and recall prompts and, where appropriate, consulting different household members to identify relevant expenditures. The consumption module records expenditure for the month preceding the interview and the three preceding months (Month L, Month L-1, Month L-2, and Month L-3) across a range of consumption categories. We construct ****total monthly household consumption expenditure**** by aggregating expenditure across food, intoxicants, restaurants, cosmetics and toiletries, power and fuel, transport, communication and information, recreation, and clothing and footwear. ****Health and education expenditures are excluded**** from this measure. The resulting variable therefore captures routine household consumption expenditure during the reference month, aggregated across the relevant expenditure categories and household members.

2.2 Household groups

We classify households into poor hand-to-mouth (P-HtM), wealthy hand-to-mouth (W-HtM), and non-hand-to-mouth (N-HtM) groups following [Gupta et al. \(2025\)](#), who adapt the liquid- and illiquid-wealth classification of [Kaplan et al. \(2014\)](#) to India. Because no

Indian household survey jointly observes income, consumption, and detailed balance sheets, they harmonize the All-India Debt and Investment Survey (AIDIS) with the Consumer Pyramids Household Survey (CPHS), using common household characteristics to predict income in AIDIS and asset and debt holdings in CPHS. For the CPHS classification, observed monthly income is combined with predicted balance sheets, with adjustments that preserve the AIDIS distributions of asset amounts and ownership. Let m_i denote net liquid wealth, a_i net illiquid wealth, and $y_i > 0$ monthly income. A household is classified as hand-to-mouth if $0 \leq m_i \leq y_i/2$, or if its negative liquid position lies sufficiently close to or beyond its assumed borrowing limit; setting that limit equal to one month's income gives the borrower condition $m_i \leq -y_i/2$. Among these households, those with $a_i \leq 0$ are P-HtM and those with $a_i > 0$ are W-HtM; all remaining households are N-HtM. The distinction therefore captures liquidity relative to income rather than poverty or total wealth. The paper additionally considers gold as illiquid wealth and divides W-HtM households by potential collateral capacity: households satisfy this condition when $m_i + 0.75g_i > y_i/2$, where g_i is gold value. This final distinction measures whether potentially pledgeable gold could bridge the liquidity shortfall, rather than observed borrowing or loan access.

3 Empirical framework to measure MPC

3.1 Moment conditions

Following [Blundell et al. \(2008\)](#), we estimate the parameters of the income and consumption process using a minimum-distance procedure based on the covariance structure of panel data. The model imposes a set of restrictions on the variances and autocovariances of income and consumption growth, which provide information on the variances of permanent and transitory income shocks and on the extent to which these shocks are transmitted to consumption. We first estimate the relevant empirical moments from the panel data and collect them in a vector m , including variances and autocovariances of income growth and covariances between income and consumption growth at different leads and lags. For a given vector of structural parameters Λ , the model generates a corresponding vector of theoretical moments $f(\Lambda)$. The parameter estimates are obtained by choosing the parameter vector that minimizes the weighted distance between the empirical and model-implied moments,

Let y_{it} and c_{it} denote log real per-capita income and consumption, residualized on period fixed effects, age group, education group, and state by weighted least squares. Income growth follows the canonical permanent-transitory process,

$$\Delta y_{it} = \zeta_{it} + \varepsilon_{it} - \varepsilon_{i,t-1}, \quad (1)$$

with ζ a permanent random-walk innovation and ε an i.i.d. transitory shock. The intuition for identification is standard: next period's income growth contains $-\varepsilon_{it}$ and nothing else from period t , so it isolates the transitory component of today's growth. The transitory and

permanent pass-throughs are

$$\psi = \frac{\text{cov}(\Delta c_{it}, \Delta y_{i,t+1})}{\text{cov}(\Delta y_{it}, \Delta y_{i,t+1})}, \quad \phi = \frac{\text{cov}(\Delta c_{it}, \sum_{k=-1}^1 \Delta y_{i,t+k})}{\text{cov}(\Delta y_{it}, \sum_{k=-1}^1 \Delta y_{i,t+k})}, \quad (2)$$

estimated as survey-weighted covariance ratios—numerically identical to an IV regression of consumption growth on income growth instrumented by its lead (Blundell et al., 2008; Kaplan and Violante, 2010). Under the null of no measurement error and process (1), ψ at the model’s decision frequency is the MPC out of a transitory shock. We require three consecutive periods per moment observation.

3.2 Temporal Aggregation

We estimate MPCs at five frequencies: monthly, quarterly, wave (four months), annual, and biennial. All five panels are constructed from the same underlying monthly household data. This allows us to examine how the measurement window affects the estimated response of consumption to income shocks.

We first regress log income and log consumption separately on demographic characteristics and year–month fixed effects. We then construct the lower-frequency panels from the resulting monthly residuals, applying the same procedure to income and consumption. Let x_{it} denote either residual for household i in month t .

Baseline aggregation. Our baseline construction, Method 1, multiplies the mean of the available monthly residuals within a period by the number of months in the target window. Let N_f denote the length of that window: 3 months for a quarter, 4 for an interview wave, 12 for a year, and 24 for a biennium. Wave windows correspond to the four recalled months within each interview wave. Let $\mathcal{T}_{ip,f}$ denote the set of observed months for household i in period p at frequency f , and let $n_{ip,f} = |\mathcal{T}_{ip,f}|$ be the number of available observations. The baseline measure is

$$X_{ip,f}^{(1)} = N_f \left(\frac{1}{n_{ip,f}} \sum_{t \in \mathcal{T}_{ip,f}} x_{it} \right). \quad (3)$$

For a complete window, this measure equals the sum of the monthly residuals. For an incomplete window, it scales the observed monthly mean to the full period length. This adjustment implicitly assigns the observed mean to the missing months; its interpretation therefore depends on how well the observed months represent the full window. Because the underlying variables are residuals in logs, the resulting measure is a scaled log residual, rather than total income or consumption in monetary units.

Sample requirements. At each frequency, we retain households with valid observations in at least three consecutive periods at that frequency. The calendar span required to satisfy this

condition increases with the length of the aggregation window. The number of households, the number of observations, and the demographic composition of the estimation sample may therefore differ across frequencies. Comparisons of estimated MPCs must account for these potential changes in sample composition.

3.2.1 Robustness

We assess the sensitivity of the baseline estimates to three alternative constructions of the income and consumption series. These alternatives vary the scaling of the residuals, the treatment of incomplete windows, and the use of overlapping rather than fixed windows.

The first alternative retains the average monthly residual within each period without multiplying it by the target window length:

$$X_{ip,f}^{(2)} = \frac{1}{n_{ip,f}} \sum_{t \in \mathcal{T}_{ip,f}} x_{it}. \quad (4)$$

At a given frequency, this measure differs from the baseline only by a constant: $X_{ip,f}^{(1)} = N_f X_{ip,f}^{(2)}$. It therefore isolates the role of this normalization.

The second alternative sums the available monthly residuals:

$$X_{ip,f}^{(3)} = \sum_{t \in \mathcal{T}_{ip,f}} x_{it}. \quad (5)$$

This construction coincides with the baseline when all N_f months are observed. When a window is incomplete, it retains only the observed contributions, whereas the baseline scales the observed mean to the full window length. This comparison therefore assesses sensitivity to the treatment of missing months. If all retained windows are complete, Methods 1 and 3 are identical.

The third alternative replaces fixed periods with backward-looking rolling windows:

$$X_{it,f}^{(4)} = \frac{1}{N_f} \sum_{j=0}^{N_f-1} x_{i,t-j}. \quad (6)$$

The quarterly measure averages the current month and the preceding two months; the wave, annual, and biennial measures use windows of four, twelve, and twenty-four months, respectively.

This construction retains a monthly time index while smoothing variation over the target window. Adjacent rolling observations share $N_f - 1$ months. The comparison with the baseline therefore changes both the placement of the windows and, when all monthly rolling observations are retained, the overlap between successive observations.

4 Results

4.1 MPCs across Time Frequencies

Table 1 reports full-sample MPC estimates using our baseline aggregation method. The permanent and transitory MPC point estimates are higher at longer measurement windows, but the increase is substantially larger for the permanent component. Estimates at monthly, quarterly, and wave frequencies are imprecise, whereas the annual and biennial estimates are more precisely estimated.

The estimated MPC out of permanent income shocks rises from 0.050 at the monthly frequency to 0.315 at the quarterly frequency and 0.504 at the wave frequency. It changes little between the wave and annual frequencies, reaching 0.518, before increasing to 0.641 at the biennial frequency. The corresponding standard errors are large at the three shortest windows: 0.440, 0.311, and 0.614, respectively. These estimates therefore provide limited evidence about the size of the permanent MPC at short horizons. By comparison, the annual and biennial estimates have standard errors of 0.065 and 0.029.

The transitory MPC is smaller and varies less across frequencies. Its point estimate remains between 0.040 and 0.050 at the monthly, quarterly, and wave frequencies, then rises to 0.093 annually and 0.095 biennially. As with the permanent component, the shorter-window estimates are imprecise. The annual and biennial transitory MPCs are more precisely estimated, with standard errors of 0.026 and 0.012, respectively.

Taken together, the point estimates suggest that the distinction between consumption responses to permanent and transitory income shocks is more pronounced at longer measurement windows. This pattern is descriptive: the standard errors reported in the table do not establish whether differences across frequencies, or between the two shock components, are statistically significant. Moreover, these estimates compare panels constructed at different frequencies; they do not trace the cumulative consumption response to a single shock. Because sample eligibility also varies with frequency, the comparison may reflect changes in sample composition as well as temporal aggregation.

Table 1: MPC Estimates across Time Frequencie

	Monthly	Quarterly	Wave	Annual	Biennial
Permanent	0.0497 (0.4399)	0.3149 (0.3108)	0.5043 (0.6136)	0.5180 (0.0647)	0.6412 (0.0291)
Transitory	0.0433 (0.2421)	0.0402 (0.0785)	0.0497 (0.1173)	0.0927 (0.0264)	0.0954 (0.0119)

Notes: The table reports marginal propensity to consume (MPC) estimates using Method 1 across alternative time-aggregation schemes. Standard errors are reported in parentheses.

4.2 MPCs by Hand-to-Mouth Status

The relationship between hand-to-mouth status and the estimated MPC depends on the measurement frequency. As seen in Table 2, at the monthly frequency, point estimates are similar across groups: 0.032 for poor hand-to-mouth (HtM) households, 0.023 for wealthy HtM households, and 0.029 for non-HtM households. Thus, the monthly estimates do not display a clear ordering in which both HtM groups have higher MPCs than non-HtM households.

Differences become more pronounced at the four-month wave frequency. The estimated MPC is 0.160 for poor HtM households, compared with 0.100 for wealthy HtM households and 0.073 for non-HtM households. The poor HtM point estimate is more than twice the non-HtM estimate, while wealthy HtM households occupy an intermediate position. The pooled estimate of 0.088 therefore conceals substantial variation in point estimates across household groups.

Table 2: BPP transitory pass-through ψ by frequency and household group

	Poor HtM	Wealthy HtM	Non-HtM	Pooled
Monthly	0.032 (0.007)	0.023 (0.006)	0.029 (0.004)	0.028 (0.003)
Wave (4m)	0.160 (0.034)	0.100 (0.013)	0.073 (0.010)	0.088 (0.007)
Biennial (24m)	0.357 (0.162)	0.257 (0.046)	0.231 (0.028)	0.248 (0.023)

Notes: Survey-weighted covariance-ratio estimates of equation (2); household-cluster bootstrap standard errors (500 replications) in parentheses. Consumption is the 11-head KVV-consistent basket, real per-capita. The wave panel averages the four recall months of each interview; the biennial panel samples months 24 apart.

This ordering persists at the biennial frequency. The estimated MPC rises to 0.357 for poor HtM households, 0.257 for wealthy HtM households, and 0.231 for non-HtM households, with a pooled estimate of 0.248. Precision differs considerably across groups, however. The standard error for poor HtM households is 0.162, compared with 0.046 for wealthy HtM households and 0.028 for non-HtM households. The higher biennial point estimate for poor HtM households should therefore be interpreted cautiously.

Overall, estimated MPCs increase with the measurement window for all three groups, a pattern that becomes clear in Figure 1. Poor HtM households have the highest point estimate at every frequency, while wealthy HtM households have higher point estimates than non-HtM households at the wave and biennial frequencies. These comparisons describe the estimated pattern; establishing differences between groups requires formal tests that account for the uncertainty in the estimates.

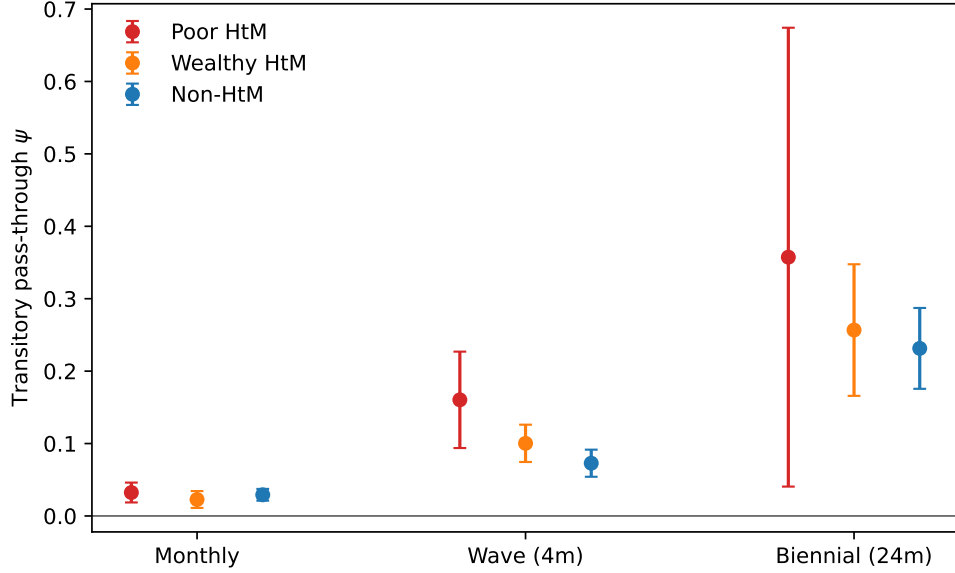


Figure 1: MPC to transitory income shocks by frequency and group

Paired comparisons across frequencies. Table 5 tests whether the increases in estimated MPCs across frequencies are statistically significant. We use 500 bootstrap replications with common household draws across frequencies, preserving the dependence between estimates when calculating the uncertainty in their differences. Each difference is defined as the shorter-frequency estimate minus the longer-frequency estimate.

For the pooled sample, the monthly MPC is 0.060 below the wave estimate ($SE = 0.008$) and 0.220 below the biennial estimate ($SE = 0.024$). The wave estimate is also 0.160 below the biennial estimate ($SE = 0.024$). All three differences are statistically significant at the 1 percent level.

The group-specific tests largely support the same pattern. Monthly MPCs are significantly below both wave and biennial MPCs for each household group. The wave-to-biennial increase is also significant for wealthy HtM and non-HtM households. For poor HtM households, however, the wave–biennial difference of -0.197 is imprecisely estimated ($SE = 0.155$, $p = 0.203$). This result does not establish that the two MPCs are equal; the estimates do not distinguish the observed increase from zero at conventional significance levels.

Overall, eleven of the twelve comparisons reject equality at the 5 percent level. The exception is the wave–biennial comparison for poor HtM households. These tests provide evidence of differences across measurement frequencies within groups; they do not test whether MPCs differ between household groups.

Comparison with PSID-based estimates. We conclude by returning to the comparison that motivated the analysis. Our pooled biennial MPC of 0.248 is higher than, but

substantially closer to, the benchmark of approximately 0.19 constructed from the PSID-based estimates of [Kaplan et al. \(2014\)](#) than is our monthly estimate of 0.028. This comparison illustrates why measurement frequency matters when interpreting differences across studies. It does not, by itself, establish that consumption responses are the same across economies: differences in samples, measurement, and estimation remain even after aligning the frequency.

Robustness to measurement and sample choices. The increase in estimated MPCs across measurement windows persists across the specifications reported in Appendix Table 6. In every reported variant, point estimates rise from monthly to wave to biennial frequency, both in the pooled sample and within each household group. The magnitude of the estimates is more sensitive to these choices than their ordering across frequencies.

Across the four consumption definitions and the unweighted specification, pooled monthly MPCs range from 0.011 to 0.028, wave MPCs from 0.064 to 0.128, and biennial MPCs from 0.186 to 0.250. Restricting consumption to the BPP nine expenditure categories produces pooled estimates close to the baseline at all three frequencies. Food-only MPCs are lower throughout, while total expenditure produces a higher wave estimate and a similar biennial estimate. Removing the weights raises the pooled wave MPC from 0.088 to 0.128, but changes the monthly and biennial estimates more modestly.

The treatment of outliers has a substantial effect on levels. Without trimming, the pooled monthly, wave, and biennial estimates are 0.018, 0.050, and 0.170, respectively. Under the fixed KVV bounds, they are 0.034, 0.148, and 0.376. The baseline estimates lie between these alternatives at each frequency. Thus, the positive cross-frequency gradient survives both trimming changes, although its magnitude depends on the treatment of extreme observations.

The pattern also survives excluding March 2020–December 2021 and, for the biennial estimates, every 24-month growth interval that overlaps this window. The resulting pooled MPCs are 0.020 monthly, 0.080 at the wave frequency, and 0.223 biennially. These estimates are below the baseline but retain the same ordering. Precision deteriorates for some household groups: in particular, the biennial poor HtM estimate has a standard error of 0.243.

Demeaning growth rates within state-by-period cells removes their state-specific period means. This adjustment produces pooled MPCs of 0.028, 0.095, and 0.258, compared with baseline estimates of 0.028, 0.088, and 0.248. The cross-frequency pattern therefore remains after this adjustment for common variation within states.

Additional checks examine incomplete waves and sample composition. Restricting the wave panel to complete four-month windows raises the pooled wave estimate to 0.146. This finding indicates that the elevated wave MPC does not depend on retaining partial waves, although the restriction changes sample selection as well as window completeness. Restricting the monthly calculation to the 5,162 households that also enter the biennial panel yields a pooled MPC of 0.0320, compared with 0.028 in the full monthly sample. The similarity of these

estimates suggests that selection into the biennial household sample does not account for the full monthly–biennial difference; it does not rule out other compositional differences.

Overall, the positive cross-frequency gradient is consistent across the reported checks. The ranking of household groups is less uniform. For example, the biennial wealthy HtM point estimate falls below the non-HtM estimate under total expenditure and when the COVID period is excluded. The robustness evidence therefore supports the ordering across measurement frequencies more consistently than a fixed ordering across household categories.

Robustness: Excluding Calendar Years 2020–2021 We next assess whether the frequency gradient depends on observations from the pandemic years. This restriction goes beyond the March 2020–December 2021 exclusion reported above: we remove every household-month dated in 2020 or 2021, every interview wave touching either year, and every biennial growth interval overlapping them. Table 7 compares the resulting estimates with the baseline.

The pooled frequency gradient remains. After the exclusion, the monthly MPC is 0.019, the wave MPC is 0.080, and the biennial MPC is 0.242. The corresponding baseline estimates are 0.028, 0.088, and 0.248. Although all three point estimates decline, the monthly–biennial gap is nearly unchanged: 0.223 after the exclusion, compared with 0.220 in the baseline.

The biennial comparison is particularly informative because the exclusion leaves only pre-pandemic variation in the identifying moments. The pooled estimate remains close to the baseline despite this restriction. Together with the continued low monthly estimate, this result indicates that the pooled cross-frequency pattern does not depend on including the pandemic years.

The exclusion has greater consequences for precision and the group-specific comparisons. The pooled biennial standard error rises from 0.023 to 0.037, and the poor HtM biennial cell becomes too imprecise to support an informative comparison. The remaining biennial point estimates also change their ordering: the wealthy HtM estimate falls to 0.174, while the non-HtM estimate rises to 0.272. The stability of the pooled result therefore does not extend to every feature of the estimated household heterogeneity.

Overall, this demanding exclusion preserves the pooled frequency gradient while reducing the information available for subgroup analysis. It complements the narrower pandemic exclusions by showing that the main pooled pattern remains when both calendar years are removed entirely.

5 Policy Transmission in a HANK Model of India

5.1 Model and calibration

We use a heterogeneous-agent New Keynesian (HANK) model calibrated to India to examine how the choice of empirical MPC target affects monetary and fiscal transmission. We compare calibrations that differ in the frequency at which the targeted MPC is measured and in how that estimate is mapped into the model’s quarterly time step.

Model environment. We solve a one-asset HANK model using the sequence-space method of [Auclert et al. \(2021\)](#). Households face idiosyncratic productivity risk and a borrowing constraint. Firms are monopolistically competitive and face nominal price rigidities. Monetary policy follows a Taylor rule, while the government issues bonds, pays lump-sum transfers, and adjusts taxes to repay debt.

The annual steady-state real interest rate is 1.5 percent, the intertemporal elasticity of substitution is 0.67, and the Frisch elasticity of labor supply is 0.37. We set the Phillips-curve slope to 0.088, following the Indian calibration based on [Gabriel et al. \(2012\)](#).

Two calibration designs. The first design targets the pooled MPC. We adjust steady-state government bond supply, B , so that the economy’s quarterly MPC matches the selected empirical target. Changing bond supply changes households’ liquid asset holdings and their consumption response to additional income. This *aggregate-liquidity design* also reveals which targets can be matched within the calibration range.

The second design targets the group-level MPCs. We introduce three permanent household types corresponding to poor HtM, wealthy HtM, and non-HtM households. Their population shares are calibrated using the estimates of [Gupta et al. \(2025\)](#). For each frequency and mapping, we calibrate the type-specific discount factors to match the corresponding group MPC targets, subject to numerical attainability. Aggregate liquid asset holdings are then determined by household saving in equilibrium.

The population shares are externally calibrated inputs and remain constant across the frequency comparisons. Changes in policy responses therefore reflect the alternative MPC targets and the resulting saving behavior, rather than changes in population weights. Because the model has only one asset, the three types capture differences in consumption responses without explicitly reproducing the liquid–illiquid portfolio distinction underlying wealthy hand-to-mouth behavior.

Idiosyncratic income risk. Both designs use the same baseline income-process parameters, $\rho_q = 0.95$ and $\sigma = 0.42$. These are maintained assumptions rather than direct esti-

Table 3: Aggregate-Liquidity Design: Policy Responses across MPC Calibrations

Mapping	Calibration	ψ pooled	Target MPC_q	Model MPC_q	Fiscal cum. 12q (%)	Monetary cum. 12q (%)	Direct share
Raw (ψ = quarterly MPC)	Monthly	0.028	0.028 [†]	0.037	+0.057	+0.068	0.25
	Wave	0.088	0.088	0.086	+0.138	+0.088	0.43
	Biennial	0.248	0.248	0.250	+0.462	+0.114	0.56
Hazard-converted	Monthly	0.028	0.081	0.079	+0.128	+0.086	0.42
	Wave	0.088	0.066	0.066	+0.106	+0.082	0.38
	Biennial	0.248	0.035 [†]	0.037	+0.057	+0.068	0.25
Published	KVW (US)	–	0.027 [†]	0.037	+0.057	+0.068	0.25

Notes: Cumulative consumption responses over twelve quarters, expressed in percent of steady-state consumption. “Direct share” is the household-block partial-equilibrium response to the equilibrium real-interest-rate path, holding other household inputs fixed, divided by the total impact consumption response. [†]The target lies below the minimum MPC attained within the calibration search, 0.037, reached at $B = 60$. Bond holdings are measured in units of quarterly GDP, so this boundary corresponds to approximately fifteen times annual GDP. Flagged targets assigned to this boundary represent the same calibrated economy.

mates from the wave panel. Holding the income process constant isolates the consequences of changing the MPC target.

We assess sensitivity to this choice using the raw wave-panel estimates, $(\rho_q, \sigma) = (0.71, 0.75)$, and a reliability-adjusted alternative, $(\rho_q, \sigma) = (0.90, 0.67)$. We also use alternative processes as sensitivity exercises, rather than assuming that all measured income variation represents risk faced by households.

Mapping empirical MPCs into quarterly targets. Let ψ_m denote an MPC estimated using a measurement window of m months. We consider two mappings:

$$\psi_q^{\text{raw}} = \psi_m, \quad (7)$$

$$\psi_q^{\text{hazard}} = 1 - (1 - \psi_m)^{3/m}. \quad (8)$$

The raw mapping assigns the empirical estimate directly to the model’s quarterly MPC. The hazard mapping interprets ψ_m as a cumulative spending response generated by a constant monthly spending hazard and converts it to a three-month response.

The hazard conversion requires an additional interpretation of the empirical estimate. It is not an aggregation identity for estimates constructed from window-averaged observations. We therefore report both mappings to show how this interpretation affects the policy conclusions. We apply the same comparison to the PSID-based benchmark.

Policy experiments. We consider a policy-rate cut of 25 basis points, expressed at an annualized rate, and a deficit-financed lump-sum transfer with an initial size of 1 percent of quarterly GDP. Both shocks have quarterly persistence of 0.6. For each calibration, we report consumption impulse responses and cumulative consumption responses over twelve quarters. Policy rules and shock specifications remain constant across the MPC calibrations.

Table 4: Three-Type Design: Policy Responses under Group-Level MPC Targets

Mapping	Calibration	B	$\beta_p/\beta_w/\beta_n$	MPC _q p/w/n	Fiscal 12q (%)	p/w/n	Monetary 12q (%)	Direct
Raw	Monthly	23.5	0.994/0.995/0.994	0.03/0.02/0.03	+0.033	+0.04/+0.02/+0.04	+0.118	0.39
	Wave	11.4	0.988/0.991/0.993	0.16/0.10/0.07	+0.095	+0.19/+0.11/+0.07	+0.133	0.33
	Biennial [†]	2.7	0.977/0.982/0.984	0.34/0.26/0.23	+0.317	+0.38/+0.34/+0.30	+0.140	0.35
	KVW	5.2	0.983/0.979/0.990	0.24/0.30/0.13	+0.236	+0.31/+0.37/+0.16	+0.144	0.28
Hazard	Monthly	12.1	0.992/0.993/0.992	0.09/0.07/0.09	+0.085	+0.11/+0.06/+0.09	+0.131	0.35
	Wave	14.7	0.990/0.993/0.994	0.12/0.08/0.05	+0.071	+0.15/+0.09/+0.05	+0.129	0.33
	Biennial	20.2	0.994/0.994/0.994	0.05/0.04/0.03	+0.038	+0.06/+0.03/+0.04	+0.122	0.37
	KVW	28.9	0.994/0.994/0.995	0.04/0.04/0.02	+0.029	+0.04/+0.06/+0.01	+0.117	0.35

Notes: Population shares are calibrated using the estimates of [Gupta et al. \(2025\)](#). Type-specific discount factors target the group MPC estimates under the raw and hazard mappings. Aggregate liquid assets, B , are determined endogenously and expressed in units of quarterly GDP. [†]At least one type-specific MPC target cannot be matched within the numerical calibration range.

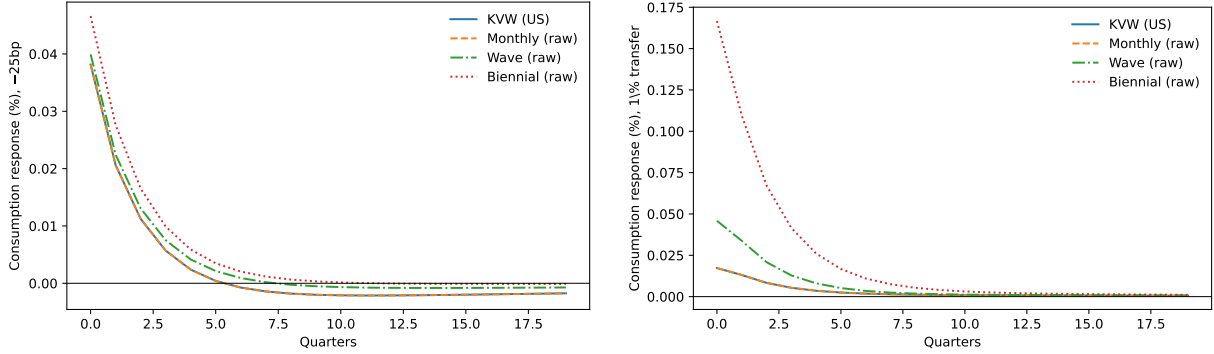


Figure 2: Consumption responses to the monetary-policy shock (left) and fiscal transfer (right) under the aggregate-liquidity design and raw MPC mapping.

5.2 Results

Fiscal transmission depends strongly on the MPC target. Tables 3 and 4 show that the consumption response to transfers is substantially more sensitive to the MPC calibration than the response to the monetary-policy shock. Under the raw mapping, the aggregate-liquidity design produces a cumulative twelve-quarter fiscal response of +0.057 percent when calibrated to the monthly estimate and +0.462 percent when calibrated to the biennial estimate, an approximately eightfold difference. Figure 2 displays the corresponding response paths.

The monthly comparison requires qualification. Its target lies below the lowest MPC attained within the calibration search, so the reported response comes from the boundary economy rather than an exact match to the monthly estimate. The comparison therefore demonstrates substantial sensitivity even over the range that the implemented model can attain.

The three-type design yields the same qualitative conclusion using group-level targets and externally calibrated population shares. Under the raw mapping, the cumulative consump-

tion response to the transfer rises from +0.033 percent under the monthly targets to +0.317 percent under the biennial targets. Monetary responses vary much less, by less than 25 percent across the reported calibrations. The sensitivity of fiscal transmission therefore persists when the model targets household heterogeneity directly. Flagged calibrations remain subject to the numerical attainability qualifications in the table.

The mapping changes the policy comparison. The hazard conversion changes both the dispersion and the ordering of the quarterly MPC targets. For estimates between zero and one, it raises a monthly estimate when converting it to a quarterly response and lowers an estimate measured over a window longer than a quarter. Consequently, the ranking of fiscal responses differs from that obtained under the raw mapping.

The relevant choice is not simply which frequency provides the most precise empirical estimate. It also concerns whether that estimate measures the consumption response that the quarterly model requires.

Monetary transmission varies less. In the aggregate-liquidity design, cumulative monetary responses range from +0.068 to +0.114 percent. Their variation is considerably smaller than that of the transfer responses, consistent with the pattern in the three-type design.

The impact decomposition helps interpret this relative stability. Changes in the household response to the real-interest-rate path are offset by changes in the contribution of other equilibrium household inputs, including income. This distinction between direct interest-rate effects and indirect equilibrium effects follows [Kaplan et al. \(2018\)](#). A small negative direct share means that the isolated interest-rate contribution reduces impact consumption; it does not imply that the total consumption response is negative. The impact decomposition also need not fully explain the cumulative twelve-quarter response.

Transfers operate differently because they change households' disposable resources directly. Their initial consumption effect depends on the MPC, while subsequent responses also depend on income, interest rates, and the taxes used to service additional debt. The cumulative fiscal response therefore reflects both household spending behavior and general-equilibrium adjustment.

The relative stability of monetary transmission is a result for the model and policy rules studied here. It is related to the aggregation benchmarks in [Werning \(2015\)](#), but those results do not establish invariance for arbitrary preferences, income risk, or asset structures. Changes in the cyclical risk, the distribution of income gains, or the available assets may alter this conclusion.

Low MPC targets place demanding restrictions on liquidity. The calibration also reveals a tension between low quarterly MPC targets and the liquid asset holdings needed to support them. In the aggregate-liquidity design, the minimum attained MPC is 0.037 at

bond holdings equal to approximately fifteen times annual GDP. Even this large liquid asset stock does not exactly match the raw monthly target.

The three-type calibration reveals a similar tension. Matching the raw monthly group targets requires aggregate liquid assets of 23.5 times quarterly GDP and discount factors close to the upper bound associated with the steady-state real interest rate. Low MPC targets therefore place substantial demands on household patience and liquidity in both designs.

Sensitivity and interpretation. The income-risk calibration affects the magnitude of the fiscal contrast. Under the raw wave-panel income process, the monthly-raw and biennial-raw cumulative responses are +0.071 and +0.986 percent, respectively, a roughly fourteenfold difference. Thus, the size of the fiscal response depends on income risk as well as the MPC target.

Local sensitivity calculations provide another measure of the stakes. Within the attainable range of the aggregate-liquidity design, changing the quarterly MPC target by 0.0035 near the lower boundary, or by 0.0187 near the biennial-raw calibration, changes the cumulative fiscal response by 10 percent. These changes can be compared with the 0.22 gap between the raw monthly and biennial targets.

Finally, the two-asset exercise reported in the replication package uses a government-spending shock, rather than the lump-sum transfer considered here, and produces a cumulative consumption response of the opposite sign. Because both the asset structure and the fiscal instrument differ, this comparison does not isolate the effect of introducing a second asset. It illustrates the importance of specifying the policy experiment alongside the model.

The central conclusion is that the choice and interpretation of the empirical MPC target can substantially change the predicted consumption response to transfers. The comparatively small variation in monetary responses is conditional on the model environment. Both findings reinforce the need to align the empirical measurement window with the consumption response being calibrated.

6 Conclusion

This paper shows that measurement frequency matters for both estimated marginal propensities to consume and their implications for policy transmission. Using a monthly household panel for India, we construct consumption and income measures at multiple frequencies and compare the resulting MPC estimates. The common data source allows us to examine differences associated with measurement windows without relying solely on comparisons across surveys or economies.

The estimates generally increase as the measurement window lengthens. In the five-frequency

analysis, the increase is larger for the permanent-income component than for the transitory component, although the shorter-frequency estimates are imprecise. In the household-group analysis, the pooled MPC rises from 0.028 monthly to 0.088 at the four-month wave frequency and 0.248 biennially. Paired bootstrap tests reject equality in eleven of the twelve cross-frequency comparisons. The exception is the wave–biennial comparison for poor HtM households, where uncertainty remains substantial.

The pooled frequency gradient persists across alternative consumption definitions, weighting and trimming rules, and adjustments for state-specific aggregate variation. It also survives removing calendar years 2020 and 2021 entirely: the pooled monthly, wave, and biennial estimates are then 0.019, 0.080, and 0.242. The pattern therefore does not depend on including the pandemic years. Household-group rankings are less stable across specifications, and some subgroup estimates become uninformative under demanding sample restrictions. The evidence for the pooled frequency gradient is consequently stronger than the evidence for a fixed ranking across household categories.

The model exercises show why these measurement differences matter. In a one-asset HANK model calibrated to India, alternative MPC targets generate substantially different consumption responses to deficit-financed transfers. This sensitivity appears both when aggregate liquidity is calibrated to the pooled MPC and when type-specific discount factors are calibrated to group-level MPCs using externally calibrated population shares. The mapping from an empirical measurement window to a quarterly model target can also change the ranking of fiscal responses. Monetary responses vary less within the model, reflecting offsetting changes in transmission channels, but this relative stability is conditional on the maintained model environment and policy rules.

The exercise does not establish that the monthly or biennial estimate is the uniquely appropriate calibration target. Estimates constructed from different windows need not measure the same response, and averaging, sampling, reporting error, and sample selection can affect their interpretation. Nor does a low empirical MPC automatically invalidate a model class. The large liquid asset holdings needed to approach the monthly targets in our calibrations instead provide an additional diagnostic: the model should be assessed against household balance sheets as well as its ability to match an MPC.

The broader implication is that an empirical MPC cannot be separated from how it is measured. Comparisons across studies require attention to measurement frequency, while quantitative policy exercises require an explicit link between the empirical estimate and the response targeted in the model. In our application, that link has substantial consequences for predicted fiscal transmission. Establishing it is therefore part of the economic analysis, rather than a preliminary normalization.

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A Additional tables

Table 5: Paired cross-frequency tests (common household draws)

Pair	Group	$\Delta\psi$	SE	z	p
Monthly – Wave	Poor HtM	-0.128***	0.036	-3.5	0.000
	Wealthy HtM	-0.078***	0.016	-4.9	$< 10^{-4}$
	Non-HtM	-0.044***	0.010	-4.3	$< 10^{-4}$
	Pooled	-0.060***	0.008	-7.1	$< 10^{-4}$
Monthly – Biennial	Poor HtM	-0.325**	0.150	-2.2	0.031
	Wealthy HtM	-0.234***	0.046	-5.1	$< 10^{-4}$
	Non-HtM	-0.202***	0.028	-7.3	$< 10^{-4}$
	Pooled	-0.220***	0.024	-9.2	$< 10^{-4}$
Wave – Biennial	Poor HtM	-0.197	0.155	-1.3	0.203
	Wealthy HtM	-0.156***	0.046	-3.4	0.001
	Non-HtM	-0.159***	0.029	-5.5	$< 10^{-4}$
	Pooled	-0.160***	0.024	-6.6	$< 10^{-4}$

Notes: differences in ψ ; 500 common-draw bootstrap replications. *, **, ***: 10, 5, 1 percent.

Table 6: Robustness: ψ across definitions, weighting, outlier rules, and sample windows

Variant	Frequency	Poor HtM	Wealthy HtM	Non-HtM	Pooled
Baseline (11 heads)	Monthly	0.032 (0.007)	0.023 (0.006)	0.029 (0.004)	0.028 (0.003)
	Wave (4m)	0.160 (0.034)	0.100 (0.013)	0.073 (0.010)	0.088 (0.007)
	Biennial (24m)	0.357 (0.162)	0.257 (0.046)	0.231 (0.028)	0.248 (0.023)
BPP 9 heads	Monthly	0.027 (0.007)	0.015 (0.006)	0.026 (0.004)	0.023 (0.003)
	Wave (4m)	0.152 (0.033)	0.101 (0.013)	0.070 (0.009)	0.086 (0.007)
	Biennial (24m)	0.295 (0.146)	0.264 (0.049)	0.234 (0.028)	0.249 (0.025)
Total expenditure	Monthly	0.031 (0.007)	0.025 (0.006)	0.029 (0.004)	0.028 (0.003)
	Wave (4m)	0.178 (0.036)	0.112 (0.014)	0.087 (0.011)	0.101 (0.008)
	Biennial (24m)	0.421 (0.132)	0.231 (0.049)	0.241 (0.030)	0.250 (0.025)
Food only	Monthly	0.011 (0.005)	0.008 (0.004)	0.012 (0.002)	0.011 (0.002)
	Wave (4m)	0.128 (0.031)	0.087 (0.013)	0.046 (0.008)	0.064 (0.007)
	Biennial (24m)	0.223 (0.141)	0.230 (0.046)	0.159 (0.026)	0.186 (0.023)
Unweighted	Monthly	0.034 (0.008)	0.018 (0.005)	0.025 (0.004)	0.024 (0.003)
	Wave (4m)	0.194 (0.033)	0.160 (0.014)	0.112 (0.010)	0.128 (0.008)
	Biennial (24m)	0.369 (0.095)	0.274 (0.034)	0.218 (0.021)	0.238 (0.017)
KVV fixed-bounds trim	Monthly	0.066 (0.030)	0.026 (0.015)	0.034 (0.010)	0.034 (0.008)
	Wave (4m)	0.240 (0.048)	0.184 (0.020)	0.119 (0.013)	0.148 (0.010)
	Biennial (24m)	0.514 (0.111)	0.416 (0.054)	0.338 (0.036)	0.376 (0.030)
No trim	Monthly	0.031 (0.006)	0.023 (0.002)	0.014 (0.002)	0.018 (0.001)
	Wave (4m)	0.096 (0.021)	0.064 (0.009)	0.040 (0.004)	0.050 (0.004)
	Biennial (24m)	0.269 (0.108)	0.215 (0.040)	0.147 (0.023)	0.170 (0.020)
Excl. COVID (2020m3–2021m12)	Monthly	0.025 (0.020)	0.017 (0.009)	0.021 (0.006)	0.020 (0.004)
	Wave (4m)	0.175 (0.045)	0.092 (0.021)	0.063 (0.013)	0.080 (0.010)
	Biennial (24m)	0.246 (0.243)	0.196 (0.045)	0.232 (0.039)	0.223 (0.028)
State $\times$ period demeaning	Monthly	0.033 (0.007)	0.022 (0.006)	0.031 (0.004)	0.028 (0.003)
	Wave (4m)	0.172 (0.035)	0.112 (0.015)	0.078 (0.009)	0.095 (0.008)
	Biennial (24m)	0.325 (0.130)	0.268 (0.045)	0.245 (0.029)	0.258 (0.022)

Notes: bootstrap standard errors in parentheses. All variants use the baseline sample, classification, and residualization unless stated. The COVID-exclusion biennial cell retains 6,494 of 9,404 moment observations after dropping pandemic-spanning growth intervals. The MA(1)-robust estimator (lead-two instrument) gives 0.0301 (s.e. 0.0096) at the monthly frequency and 0.1737 (0.0314) at the wave frequency; at the biennial frequency the lead-two identifying covariance is positive and near zero, so the estimator is uninformative there. Non-duplicator staircase: 0.026/0.070/0.227 (Section ??).

Table 7: MPC Estimates Excluding Calendar Years 2020–2021

Frequency	Sample	Poor HtM	Wealthy HtM	Non-HtM	Pooled
Monthly	Baseline (2014–2023)	0.032 (0.007)	0.023 (0.006)	0.029 (0.004)	0.028 (0.003)
	Excl. 2020–2021	0.024 (0.020)	0.017 (0.010)	0.019 (0.006)	0.019 (0.005)
Wave (4m)	Baseline (2014–2023)	0.160 (0.034)	0.100 (0.013)	0.073 (0.010)	0.088 (0.007)
	Excl. 2020–2021	0.175 (0.044)	0.092 (0.023)	0.063 (0.013)	0.080 (0.010)
Biennial (24m)	Baseline (2014–2023)	0.357 (0.162)	0.257 (0.046)	0.231 (0.028)	0.248 (0.023)
	Excl. 2020–2021	–	0.174 (0.058)	0.272 (0.046)	0.242 (0.037)

Notes: Bootstrap standard errors in parentheses, based on 500 household-cluster replications. The exclusion removes all household-months dated in 2020 or 2021, interview waves touching those years, and biennial growth intervals overlapping them. Biennial identification therefore uses only pre-2020 variation, with one eligible 2014 → 2016 → 2018 moment triple per household and 3,854 observations. The poor HtM biennial cell is suppressed because it contains only 224 observations and has a bootstrap standard error greater than one.

Table 8: MPC estimation across methods and frequencies

	Quarterly	Wave	Annual	Biennial
<i>Panel A: Method 1</i>				
Permanent	0.3149 (0.3108)	0.5043 (0.6136)	0.5180 (0.0647)	0.6412 (0.0291)
Transitory	0.0402 (0.0785)	0.0497 (0.1173)	0.0927 (0.0264)	0.0954 (0.0119)
<i>Panel B: Method 2</i>				
Permanent	0.3149 (2.7969)	0.5043 (5.5226)	0.5180 (9.3156)	0.6412 (16.7371)
Transitory	0.0402 (0.7068)	0.0497 (1.0554)	0.0927 (3.8065)	0.0954 (6.8575)
<i>Panel C: Method 3</i>				
Permanent	0.4640 (0.4982)	0.6666 (0.6390)	0.8928 (0.1414)	0.7121 (0.0364)
Transitory	0.0483 (0.1109)	0.0913 (0.1192)	−0.0293 (0.1450)	0.2371 (0.0162)
<i>Panel D: Method 4</i>				
Permanent	0.0544 (0.9742)	0.0000 (0.0000)	0.0968 (7.5641)	0.1033 (17.0301)
Transitory	0.0804 (3.0852)	0.0000 (0.0000)	0.1622 (20.3812)	0.1650 (44.3354)

Notes: Standard errors are reported in parentheses. Estimates are reported for alternative time-aggregation schemes.